```
import numpy as np
import pandas as pd
```

```
df=pd.read_csv("/content/full_dataset.csv")
```

```
df.head()
```

	label	acc_x_0	acc_x_1	acc_x_2	acc_x_3	acc_x_4	acc_x_5	acc_x_6	acc_x_7	acc_x_8	...	gy_z_150	gy_z_151	gy_z_152
0	idle	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	...	1.6440	1.6856	1.5975
1	fall	-0.0008	0.0181	0.1298	0.1815	0.1867	0.0526	0.0188	0.0228	0.3152	...	1.2886	1.2177	1.3184
2	step	0.0200	0.0200	0.0276	0.0201	0.0000	0.0000	0.0000	0.0107	0.0291	...	-0.5565	-0.1801	-0.2183
3	motion	-0.0348	-0.0104	0.0309	0.0373	0.0498	0.0600	0.0984	0.0810	0.0559	...	-0.5500	-0.5500	-0.2313
4	step	0.0200	0.0172	0.0100	0.0020	-0.0099	0.0024	0.0155	0.0357	0.0200	...	-0.5644	-0.5155	-0.5806

5 rows × 961 columns

```
df.isnull().count()
```

	0
label	160
acc_x_0	160
acc_x_1	160
acc_x_2	160
acc_x_3	160
...	...
gy_z_155	160
gy_z_156	160
gy_z_157	160
gy_z_158	160
gy_z_159	160

961 rows × 1 columns

dtype: int64

```
df.describe()
```

	acc_x_0	acc_x_1	acc_x_2	acc_x_3	acc_x_4	acc_x_5	acc_x_6	acc_x_7	acc_x_8	acc_x_9	...
count	160.000000	160.000000	160.000000	160.000000	160.000000	160.000000	160.000000	160.000000	160.000000	160.000000	...
mean	0.019386	0.023758	0.029116	0.029559	0.030321	0.025803	0.024854	0.029618	0.040863	-0.002831	...
std	0.168756	0.204312	0.207765	0.209399	0.215238	0.197174	0.210010	0.253022	0.258793	0.341709	...
min	-1.170100	-1.130400	-1.110000	-1.104000	-1.109800	-1.100000	-1.100000	-1.084400	-1.059700	-3.494800	...
25%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	-0.000050	0.000000	0.000000	0.000000	...
50%	0.020000	0.018000	0.019700	0.014650	0.013350	0.020000	0.016850	0.014000	0.019850	0.014300	...
75%	0.027950	0.025175	0.030000	0.029925	0.023400	0.025600	0.030000	0.030000	0.029550	0.022875	...
max	1.364100	2.032800	2.102900	2.143400	2.192600	1.941600	2.115700	1.966300	2.112700	1.657500	...

8 rows × 960 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 160 entries, 0 to 159
Columns: 961 entries, label to gy_z_159
dtypes: float64(960), object(1)
memory usage: 1.2+ MB
```

```
import sklearn.model_selection
from sklearn.preprocessing import LabelEncoder
```

```
label = ['idle','fall','step','motion',]
```

```
label_encoder = LabelEncoder()
encoded_labels = label_encoder.fit_transform(label)
```

```
print(encoded_labels) # Output: [2 1 0 1 2] (or similar, order depends on internal sort)
print(label_encoder.classes_)
```

```
[1 0 3 2]
['fall' 'idle' 'motion' 'step']
```

```
df['encoded_label'] = label_encoder.fit_transform(df['label'])
display(df.head())
```

	label	acc_x_0	acc_x_1	acc_x_2	acc_x_3	acc_x_4	acc_x_5	acc_x_6	acc_x_7	acc_x_8	...	gy_z_151	gy_z_152	gy_z_153
0	idle	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	...	1.6856	1.5975	1.4654
1	fall	-0.0008	0.0181	0.1298	0.1815	0.1867	0.0526	0.0188	0.0228	0.3152	...	1.2177	1.3184	1.1846
2	step	0.0200	0.0200	0.0276	0.0201	0.0000	0.0000	0.0000	0.0107	0.0291	...	-0.1801	-0.2183	-0.3190
3	motion	-0.0348	-0.0104	0.0309	0.0373	0.0498	0.0600	0.0984	0.0810	0.0559	...	-0.5500	-0.2313	-0.1530
4	step	0.0200	0.0172	0.0100	0.0020	-0.0099	0.0024	0.0155	0.0357	0.0200	...	-0.5155	-0.5806	-0.6057

5 rows × 962 columns

The `encoded_label` column now contains numerical representations of your original labels. Here's how the original labels are mapped to the encoded numbers:

```
for i, label_class in enumerate(label_encoder.classes_):
    print(f"{label_class}: {i}")
```

```
fall: 0
idle: 1
motion: 2
step: 3
```

```
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LinearRegression, LogisticRegression, ElasticNet
from sklearn.ensemble import RandomForestClassifier
```

```
target = df['label']
y = df['encoded_label']
x = df.drop(['label', 'encoded_label'], axis=1)
```

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.33, random_state=42)

# Impute missing values with the mean
from sklearn.impute import SimpleImputer
imputer = SimpleImputer(strategy='mean')
x_train = imputer.fit_transform(x_train)
x_test = imputer.transform(x_test)
```

```
Logreg=LogisticRegression
```

```
Logreg_model = Logreg()
Logreg_model.fit(x_train, y_train)
```

```
/usr/local/lib/python3.12/dist-packages/sklearn/linear_model/_logistic.py:465: ConvergenceWarning: lbfgs failed to converge
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
  n_iter_i = _check_optimize_result(
```

```
▾ LogisticRegression
LogisticRegression()
```

```
y_pred = Logreg_model.predict(x_test)
print("Accuracy:", accuracy_score(y_test, y_pred)*100)
```

```
Accuracy: 47.16981132075472
```

```
model = ElasticNet(alpha=1.0, l1_ratio=0.5, random_state=0)
```

```
model.fit(x_train, y_train)
```

```
▾ ElasticNet
ElasticNet(random_state=0)
```

```
y_pred = model.predict(x_test)
```

```
print(f"R^2 score on test data: {model.score(x_test, y_test)}")
```

```
R^2 score on test data: 0.3249235054767593
```

```
import numpy as np

# Round the continuous predictions to the nearest integer
y_pred_discrete = np.round(y_pred)

# Ensure predictions are within the valid range of encoded labels (0, 1, 2, 3)
# np.clip(value, min, max) ensures values are not below min or above max
y_pred_discrete = np.clip(y_pred_discrete, a_min=0, a_max=len(label_encoder.classes_)-1)

print("Accuracy:", accuracy_score(y_test, y_pred_discrete)*100)
```

```
Accuracy: 39.62264150943396
```

Let's try training a `RandomForestClassifier` to see if we can get better performance. Random Forests are powerful ensemble learning methods that can handle high-dimensional data and are less sensitive to feature scaling.

```
# Initialize and train a RandomForestClassifier
rf_model = RandomForestClassifier(random_state=42 , n_estimators = 8000)
rf_model.fit(x_train, y_train)

# Make predictions on the test set
y_pred_rf = rf_model.predict(x_test)

# Evaluate the model's accuracy
accuracy_rf = accuracy_score(y_test, y_pred_rf) * 100
print(f"Random Forest Classifier Accuracy: {accuracy_rf:.2f}%")
```

```
Random Forest Classifier Accuracy: 86.79%
```

```
from sklearn.svm import SVC
# Use SVC (Classification) instead of SVR (Regression) for categorical labels
classifier = SVC(kernel='rbf', C=100, gamma='scale')
classifier.fit(x_train, y_train)

# 5. Predict
y_pred = classifier.predict(x_test)
accuracy_svc = accuracy_score(y_test, y_pred) * 100
print(f"SVM Classifier Accuracy: {accuracy_svc:.2f}%")
```

```
SVM Classifier Accuracy: 49.06%
```

```
from sklearn.model_selection import RandomizedSearchCV, KFold

# 1. Define the parameter grid to search
param_dist = {
    'n_estimators': [100, 500, 1000],
    'max_depth': [None, 10, 20, 30],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4]
}

kfold = KFold(n_splits=5, shuffle=True, random_state=42)

# 2. Instantiate RandomizedSearchCV with param_distributions
rcv = RandomizedSearchCV(
    estimator=rf_model,
    param_distributions=param_dist,
    scoring='accuracy',
    refit=True,
    cv=kfold,
    verbose=2,
    n_iter=5,
    random_state=42
)

# 3. Fit the search (this might take a moment)
rcv.fit(x_train, y_train)
print("Best parameters:", rcv.best_params_)
print("Best score:", rcv.best_score_)
```

```
Fitting 5 folds for each of 5 candidates, totalling 25 fits
[CV] END max_depth=20, min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total time=   4.4s
[CV] END max_depth=20, min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total time=   2.6s
[CV] END max_depth=20, min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total time=   2.5s
[CV] END max_depth=20, min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total time=   2.6s
[CV] END max_depth=20, min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total time=   2.6s
[CV] END max_depth=None, min_samples_leaf=2, min_samples_split=2, n_estimators=500; total time=   2.2s
[CV] END max_depth=None, min_samples_leaf=2, min_samples_split=2, n_estimators=500; total time=   1.9s
[CV] END max_depth=None, min_samples_leaf=2, min_samples_split=2, n_estimators=500; total time=   2.0s
[CV] END max_depth=None, min_samples_leaf=2, min_samples_split=2, n_estimators=500; total time=   1.4s
[CV] END max_depth=None, min_samples_leaf=2, min_samples_split=2, n_estimators=500; total time=   1.3s
[CV] END max_depth=None, min_samples_leaf=1, min_samples_split=5, n_estimators=500; total time=   1.7s
[CV] END max_depth=None, min_samples_leaf=1, min_samples_split=5, n_estimators=500; total time=   1.5s
[CV] END max_depth=None, min_samples_leaf=1, min_samples_split=5, n_estimators=500; total time=   2.0s
[CV] END max_depth=None, min_samples_leaf=1, min_samples_split=5, n_estimators=500; total time=   2.6s
[CV] END max_depth=None, min_samples_leaf=1, min_samples_split=5, n_estimators=500; total time=   1.5s
[CV] END max_depth=30, min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total time=   2.6s
[CV] END max_depth=30, min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total time=   2.6s
[CV] END max_depth=30, min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total time=   2.7s
[CV] END max_depth=30, min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total time=   3.7s
[CV] END max_depth=30, min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total time=   2.8s
[CV] END max_depth=20, min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total time=   2.7s
[CV] END max_depth=20, min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total time=   2.9s
[CV] END max_depth=20, min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total time=   3.2s
[CV] END max_depth=20, min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total time=   3.2s
[CV] END max_depth=20, min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total time=   2.9s
Best parameters: {'n_estimators': 500, 'min_samples_split': 2, 'min_samples_leaf': 2, 'max_depth': None}
Best score: 0.8887445887445887
```

```
df.shape
```

```
(160, 962)
```

```
# evaluation_metrics.py

import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import (
    confusion_matrix,
    ConfusionMatrixDisplay,
    classification_report,
    roc_curve,
    auc,
    RocCurveDisplay
)
from sklearn.preprocessing import label_binarize


def evaluate_with_plots(model, X_test, y_test, label_encoder):
    """
    Detailed evaluation with plots and metrics
    """

    y_pred = model.predict(X_test)

    print("\n" + "="*50)
    print("CLASSIFICATION REPORT")
    print("="*50)
    print(classification_report(y_test, y_pred, target_names=label_encoder.classes_))

    # ---------------- Confusion Matrix ----------------
    cm = confusion_matrix(y_test, y_pred)
    disp = ConfusionMatrixDisplay(confusion_matrix=cm,
                                  display_labels=label_encoder.classes_)
    disp.plot(cmap="Blues")
    plt.title("Confusion Matrix")
    plt.show()

    # ---------------- ROC Curve (Multiclass) ----------------
    y_test_bin = label_binarize(y_test, classes=range(len(label_encoder.classes_)))
    y_score = model.predict_proba(X_test)

    fpr = dict()
    tpr = dict()
    roc_auc = dict()

    for i in range(len(label_encoder.classes_)):
        fpr[i], tpr[i], _ = roc_curve(y_test_bin[:, i], y_score[:, i])
        roc_auc[i] = auc(fpr[i], tpr[i])

    # Plot ROC for each class
    plt.figure(figsize=(8, 6))
    for i, class_name in enumerate(label_encoder.classes_):
        plt.plot(fpr[i], tpr[i], label=f"{class_name} (AUC = {roc_auc[i]:.2f})")

    plt.plot([0, 1], [0, 1], linestyle="--")
    plt.title("ROC Curve (Multiclass)")
    plt.xlabel("False Positive Rate")
    plt.ylabel("True Positive Rate")
    plt.legend()
    plt.show()

    print("\nROC AUC Scores per class:")
    for i, class_name in enumerate(label_encoder.classes_):
        print(f"{class_name}: {roc_auc[i]:.3f}")
```

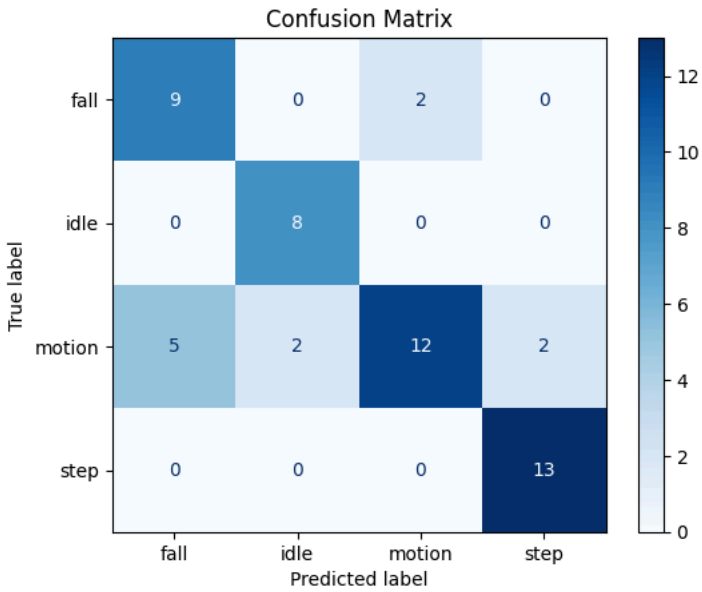
```
best_model = rcv.best_estimator_
```

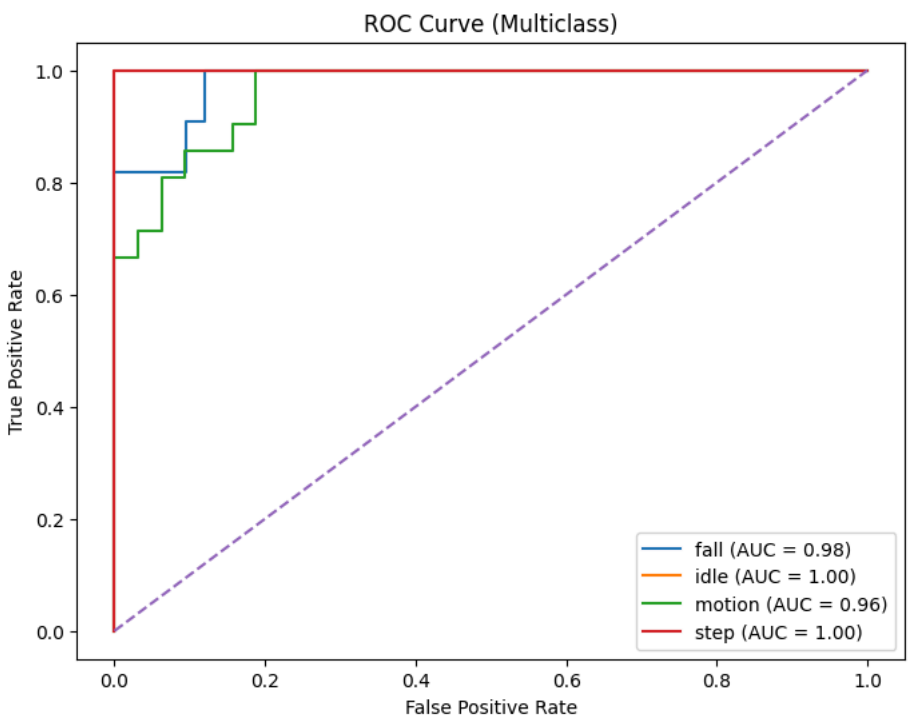
```
evaluate_with_plots(best_model, x_test, y_test, label_encoder)
```

```
==================================================
CLASSIFICATION REPORT
==================================================
              precision    recall  f1-score   support

        fall       0.64      0.82      0.72        11
        idle       0.80      1.00      0.89         8
      motion       0.86      0.57      0.69        21
        step       0.87      1.00      0.93        13

    accuracy                           0.79        53
   macro avg       0.79      0.85      0.81        53
weighted avg       0.81      0.79      0.78        53
```






```
ROC AUC Scores per class:
fall: 0.981
idle: 1.000
motion: 0.963
step: 1.000
```